Assessment of the Challenges in Creating High-Quality, Semantically Rich Representations from Aerial Data

CS592: Autonomous Drones

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***Abstract*—This report identifies a fundamental system paradox in Unmanned Aerial Vehicle (UAV) spatial AI: the mutually exclusive trajectory requirements for optimal 3D geometric reconstruction and zero-shot semantic understanding. By evaluating state-of-the-art 3D Gaussian Splatting (3DGS) alongside Vision-Language Models (VLMs) across multi-tier drone trajectories on six UAVScenes sequences, we demonstrate that low-altitude flights (< 30m) maximize geometric fidelity (PSNR up to 36.1 dB) but severely restrict semantic context, yielding poor zero-shot accuracy (as low as 2.3%). Conversely, high-altitude flights (> 60m) provide the holistic context required for accurate VLM semantics (up to 25.1% accuracy) but cause catastrophic geometric degradation and Structure-from-Motion (SfM) registration failures across multiple sequences. Furthermore, we expose a benchmark fallacy where zero-shot foundation models (SAM 2) score artificially low (Boundary F1 < 0.14) against coarse human annotations due to precision mismatch. These findings dictate that future autonomous active perception systems must physically decouple geometry and semantics using distance-dependent, multi-tier flight profiles.**

This report also presents a systematic empirical study isolating the direct impact of motion-blurred training data on 3DGS reconstruction fidelity. We use a reference-free, Fast Fourier Transform (FFT)-based blur quantification pipeline to progressively filter a real-world dataset into thresholded training subsets. Our quantitative evaluation reveals that while 3DGS exhibits robustness to the removal of severely blurred images, the retention of moderately blurred images actively bottlenecks convergence by introducing conflicting views. By systematically eliminating these artifacts, we demonstrate that the perceptual and structural gains of utilizing exclusively sharp, high-frequency images can outweigh the geometric penalties of reduced multi-view density. The highest rendering fidelity was achieved at our strictest evaluation threshold, underscoring that curated data quality supersedes raw data volume. These findings highlight the necessity of blur-aware data curation for aerial data.

***Index Terms*—Autonomous Drones, UAV, Robotics, Computer Vision, 3D Gaussian Splatting, Vision-Language Models**

I. Introduction

The emergence of open-vocabulary 3D scene understanding has been driven by distilling high-dimensional Vision-Language Model (VLM) features into spatial representations. Architectures such as LERF [3] and LangSplat [6] successfully embed CLIP [8] features into Neural Radiance Fields and 3D Gaussian Splatting (3DGS) [2] primitives, respectively, enabling natural language queries over reconstructed 3D maps. The recently released LangSplatV2 [7] further advances this paradigm to over 450 FPS rendering via high-dimensional feature compression. These systems have demonstrated impressive capabilities for indoor and ground-level outdoor scenes, where a single camera trajectory can simultaneously collect the micro-geometric parallax required for dense 3D reconstruction and the macro-contextual views required for accurate semantic grounding.

However, the application of these pipelines to outdoor UAV data introduces a fundamental conflict that has not been systematically characterized. Firstly, UAV platforms necessarily traverse a wide altitude range during a single mission, from low-altitude texture collection to high-altitude surveying passes. At low altitudes, high angular parallax enables accurate Structure-from-Motion (SfM) initialization and 3DGS optimization, yielding sharp novel view synthesis. Yet at low altitude, the camera's fixed receptive field captures only local surface texture: a CLIP ViT-B/32 encoder processing a 2448×2048 aerial image at 20m altitude sees only a small patch of rooftop or road surface with no recognizable object boundary. Conversely, at high altitude, the holistic bird's-eye perspective provides the full object context that VLMs were pretrained on, improving zero-shot recognition accuracy—but the loss of angular parallax causes catastrophic SfM registration failures and severe photometric degradation in 3DGS reconstruction.

Secondly, outdoor UAV data frequently exhibits motion blur induced by sudden platform movements and rapid perspective shifts, an effect exacerbated in low-light environments. These kinematics are intrinsic to aerial systems subjected to external aerodynamic disturbances, such as wind, where relative perspective changes are magnified by extended standoff distances. Consequently, the presence of blurred frames degrades the fidelity of the resulting Gaussian splats, as the 3DGS optimization framework relies on gradient descent to minimize

photometric loss against 2D source images.

This report identifies, quantifies, and mechanistically explains the geometry-semantics paradox and conducts a quantitative study on the effects of motion blur on 3DGS fidelity. Part one of our report, covering the preliminary testing of LangSplatV2 on outdoor UAV imagery [12] confirms the pipeline's failure to generalize across altitude scales, motivating a controlled altitude-stratified evaluation. Using the UAVScenes dataset [11], we partition raw UAV trajectories into altitude tiers and conduct parallel evaluations of (1) 3DGS geometric fidelity via novel view synthesis metrics, (2) zero-shot semantic accuracy of CLIP [8] against labeled class maps, and (3) boundary precision of Segment Anything Model 2 (SAM 2) [9] against human-annotated ground truth. Our findings establish three key results: (i) low-altitude flights achieve optimal 3DGS fidelity (PSNR up to 36.1 dB) but catastrophically degrade VLM semantic accuracy; (ii) high-altitude flights more than double zero-shot accuracy while causing systematic SfM failures; and (iii) standard Boundary F1 metrics systematically underestimate foundation model performance due to annotation granularity mismatch. Part two of this report addresses the degradation of 3DGS [2] caused by motion blur. This study demonstrates that frequency-domain quantification via a Fast Fourier Transform (FFT)-based algorithm [5] effectively ranks images by blur severity. This automated sorting correlates strongly with human perceptual assessments. Leveraging this quantification, we introduce a threshold-based rejection strategy to filter degraded frames. Experimental results indicate that discarding blurred images across varying thresholds significantly enhances the reconstruction fidelity of the resulting 3DGS models, yielding improvements of up to 1 dB in PSNR, a 0.04 increase in SSIM, and a 0.08 reduction in LPIPS.

II. Related Work

A. 3D Gaussian Splatting

3DGS [2] represents scenes as collections of anisotropic 3D Gaussians optimized via differentiable tile-based rasterization. Each Gaussian is parameterized by a 3D mean position, a full 3D covariance matrix, an opacity value, and spherical harmonic color coefficients. Initialized from a sparse SfM point cloud, the Gaussians are iteratively densified and pruned to minimize a combined photometric and structural similarity loss. The pipeline achieves real-time rendering with high fidelity on ground-level captures, but is critically sensitive to the quality of the SfM initialization—a dependency that proves fragile under aerial nadir-view geometry where inter-frame parallax collapses at high altitude.

B. Language-Embedded Radiance Fields

LERF [3] distills multi-scale CLIP feature vectors into a continuous NeRF volume, enabling open-vocabulary 3D relevancy queries at arbitrary scale. LangSplat [6] improves rendering efficiency by associating CLIP features directly with 3DGS Gaussians, with each Gaussian encoding language embeddings at three spatial scales via a learned autoencoder. LangSplatV2 [7] extends this to 450+ FPS through high-dimensional feature compression with negligible quality loss. These systems share a critical assumption: that the training trajectory provides semantically coherent views across all frames—an assumption that fails when altitude-induced scale variation creates an irreconcilable conflict between geometric and semantic requirements.

C. Vision-Language Models for Aerial Perception

CLIP [8] encodes images and text into a shared embedding space via contrastive pretraining on web-scale image-caption pairs. Its training distribution is heavily biased toward terrestrial, human-scale imagery at human eye level. UAV nadir views—particularly at low altitude where only surface texture is visible without identifiable object context—represent a significant domain shift. SAM 2 [9] generalizes the Segment Anything framework to video, producing high-fidelity instance masks with fine-grained boundary precision that far exceeds typical human polygon annotations in UAV datasets. This granularity mismatch is the root cause of the benchmark artifact we characterize in Section V-D.

D. Image Quality Assessment (IQA)

Popular methods for image quality assessment (IQA) include the Variance of the Laplacian [4] and BRISQUE (Blind/Referenceless Image Spatial Quality Evaluator) [1]. However, empirical evaluations indicate that these techniques fail to produce reliable metrics for outdoor UAV datasets. The Variance of the Laplacian evaluates sharpness by detecting spatial-domain edges against softer backgrounds; however, outdoor scenes captured at extended standoff distances often lack distinct structural edges and are instead characterized by complex textural patches. These textures contain dense, high-frequency 'micro-edges' that render Laplacian-based variance ineffective due to its high susceptibility to noise in the spatial domain. Furthermore, BRISQUE relies on a Support Vector Regression (SVR) model trained on the natural scene statistics of the LIVE IQA Database [10]. Because this database consists primarily of terrestrial imagery and does not account for the statistical distribution of aerial perspectives, BRISQUE generalizes poorly when assessing the UAVScenes dataset.

Part I: Language Fields for Drone

III. Methodology

A. System Architecture

The evaluation pipeline is divided into two decoupled tracks, as illustrated in Fig. 1. The geometric track processes altitude-stratified image subsets through COLMAP SfM for sparse point cloud initialization, followed by 3DGS optimization for 30,000 iterations with the original hyperparameters of [2]. Where available, LiDAR point clouds from UAVScenes are used to supplement the SfM initialization. Novel view synthesis quality is quantified on held-out test images. The semantic track processes the same altitude-stratified image subsets through SAM 2 for dense instance segmentation, followed by CLIP (ViT-B/32) for open-vocabulary feature

embedding. Predicted semantic labels are matched against ground-truth class ID maps. The two tracks share no parameters, enabling independent quantification of geometric and semantic performance across altitude tiers.

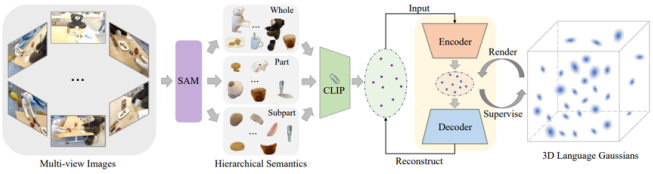


Fig. 1. Architecture of LangSplat v2. The system utilizes a combined features of SAM and CLIP to construct 3D Gaussian Splats for language field

B. Altitude Stratification

A central challenge in UAV-based evaluation is that real mission trajectories do not record imagery at fixed altitudes: the drone continuously varies altitude during a single sequence. Rather than discarding cross-altitude footage, we dynamically partition each image stream using GPS-derived altitude thresholds calibrated per sequence:

- **Low Tier ($<$ 28–35m):** High angular parallax baseline; restricted field of view; fine-grained surface texture dominates the image frame without object-level context.
- **Mid Tier (28–63m):** Transitional zone where neither geometric parallax nor semantic context is fully maximized. This tier proves to be the worst of both worlds.
- **High Tier ($>$ 58–63m):** Near-nadir perspective; low parallax; full objects and their environmental context are captured within a single image frame.

Exact altitude thresholds vary slightly per sequence due to the natural altitude distributions of each trajectory (Table I). Image overlap rates (80% frontal, 70% lateral) and lighting conditions were strictly controlled across all strata to isolate altitude as the primary independent variable.

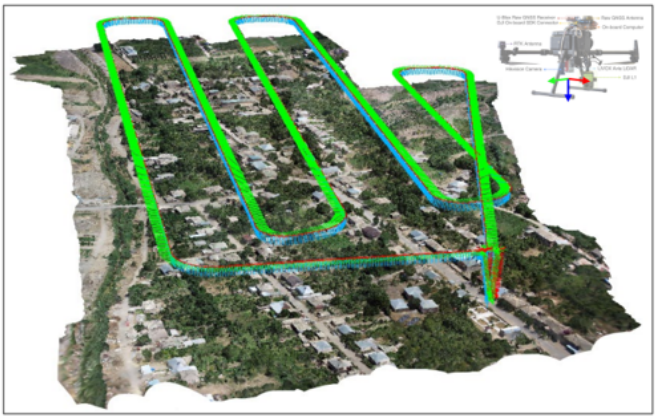


Fig. 2. Overview of the UAVScenes dataset showing representative imagery across altitude tiers and the corresponding semantic annotation layers used for ground-truth evaluation.

IV. Experimental Setup

A. Dataset

All experiments use the UAVScenes dataset [11], a multi-modal UAV aerial benchmark providing 5 MP imagery (2448×2048), LiDAR point clouds, 6-DoF camera poses, and manually-labeled semantic annotations as shown in 2. Six sequences are evaluated, spanning a range of flight patterns and altitude profiles:

TABLE I
UAVScenes Sequences Used in Evaluation

Sequence	Low / Mid / High Threshold	3DGS Status
HKisland01	28.4m / 28.4-58.1m / 58.1m	Stable
HKisland02	28.1m / 28.1-58.7m / 58.7m	Stable
HKisland03	27.1m / 27.1-56.4m / 56.4m	Stable
HKisland_GNSS01	29.0m / 29.0-59.8m / 59.8m	High: SfM failure
HKisland_GNSS02	32.3m / 32.3-63.3m / 63.3m	Mid, High: SfM failure
HKisland_GNSS03	35.0m / 35.0-60.0m / 60.0m	Semantic eval only

The GNSS sequences exhibit higher altitude variance and more aggressive heading changes during ascent and descent, which increase the frequency of SfM registration failures at higher altitude tiers. HKisland_GNSS03 is used exclusively for SAM and CLIP evaluation due to incomplete 3DGS convergence across tiers.

B. Evaluation Metrics

Geometric fidelity is assessed using three standard Novel View Synthesis (NVS) metrics: Peak Signal-to-Noise Ratio (PSNR, dB), Structural Similarity Index (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS) [13]. PSNR and SSIM measure pixel-level photometric accuracy and structural preservation, respectively, while LPIPS captures perceptual quality as assessed by a pretrained deep feature network. Higher PSNR and SSIM indicate better reconstruction; lower LPIPS indicates fewer perceptual artifacts.

Semantic accuracy is measured by projecting SAM 2 instance masks and CLIP-predicted class labels onto ground-truth semantic ID maps. For each image, the predicted class for each SAM 2 region is determined by the maximum cosine similarity between the CLIP embedding of the region's image crop and the text embeddings of all ground-truth class names in the dataset vocabulary. Instance-level accuracy is the fraction of ground-truth objects whose predicted label matches the annotation.

Boundary precision is evaluated using the Boundary F1 score, which measures the spatial overlap between predicted segment edges and ground-truth polygon annotations at a fixed pixel-distance tolerance. This metric directly exposes the annotation granularity mismatch discussed in Section V-D, as it penalizes fine-grained boundaries that do not align with coarse human-drawn polygons.

V. Results

A. Geometric Reconstruction Collapse

Table II reports 3DGS fidelity for all primary HKisland sequences. The pattern is unambiguous and consistent across

all three stable sequences: low and mid tiers achieve PSNR above 31 dB in every case, peaking at 36.1 dB for the HKisland02 low-only split, while high-only splits suffer catastrophic degradation to 25–27 dB across the board. The SSIM degradation from low to high tier (from ∼0.96 to ∼0.93) further confirms structural information loss, and the LPIPS increase from ∼0.046 to ∼0.071 indicates significant perceptual quality deterioration. Critically, the GNSS sequences exhibit outright SfM registration failures at high altitude, precluding any 3DGS training for those subsets.

TABLE II
3DGS GEOMETRIC FIDELITY BY ALTITUDE TIER

Sequence	Split	PSNR	SSIM	LPIPS
HKisland01	Low only	31.37	0.957	0.048
HKisland01	Mid only	31.01	0.962	0.050
HKisland01	High only	26.12	0.923	0.076
HKisland02	Low only	35.18	0.964	0.044
HKisland02	Mid only	34.24	0.967	0.046
HKisland02	High only	25.83	0.929	0.069
HKisland03	Low only	34.70	0.963	0.045
HKisland03	Mid only	34.49	0.965	0.048
HKisland03	High only	26.62	0.930	0.069
HKisland_GNSS01	High only	*FAILED SfM*		
HKisland_GNSS02	Mid only	*FAILED SfM*		
HKisland_GNSS02	High only	*FAILED SfM*		
3-seq. mean (Low)		*33.75*	*0.961*	*0.046*
3-seq. mean (Mid)		*33.25*	*0.963*	*0.048*
3-seq. mean (High)		*26.19*	*0.927*	*0.071*

The mechanistic cause is angular parallax collapse. At altitudes exceeding 58m over urban terrain, the baseline between adjacent image frames decreases relative to the depth of observed structures, causing COLMAP's feature matching to produce degenerate or near-planar point clouds. 3DGS is then initialized from a structurally incorrect scaffold, and the subsequent Gaussian densification diverges: rather than converging to geometry-faithful Gaussians, the optimizer places large, opaque Gaussians at incorrect depths to minimize photometric loss without recovering structural integrity. This manifests as the characteristic floater artifacts visible in Fig. 3.

Table III shows that mixed-altitude training does not recover performance: the Low+High and Mid+High combinations yield PSNR values 3–5 dB below the Low+Mid combination, confirming that high-altitude images actively degrade reconstruction quality when mixed with geometrically informative low-altitude frames. This rules out naive image pooling as a mitigation strategy and reinforces the need for physically decoupled flight profiles.

B. Semantic Understanding Paradox

Table IV reports CLIP zero-shot accuracy across all evaluated sequences. The altitude-accuracy relationship is consistent: accuracy increases with altitude in all but one case (HKisland_GNSS01, where mid and high tiers produce near-equal accuracy at 21.1% and 21.2%, respectively). The mean

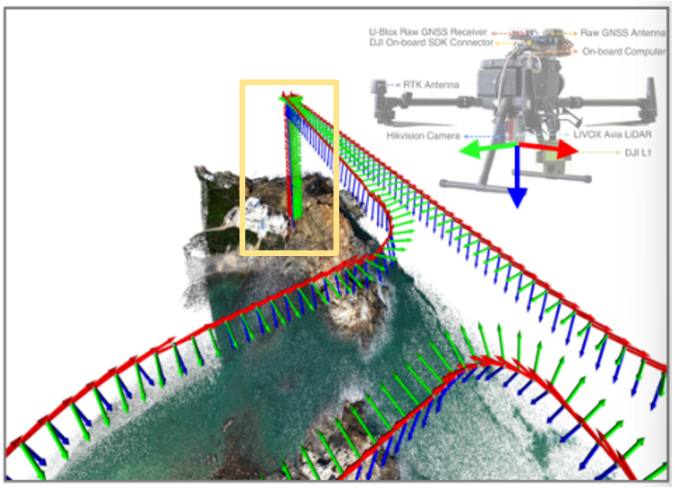


Fig. 3. Overview of the selected Area

TABLE III
3DGS FIDELITY FOR COMBINED ALTITUDE SPLITS (3-SEQUENCE MEAN, HKISLAND01–03)

Split Combination	PSNR (avg)	SSIM (avg)	LPIPS (avg)
Low + Mid	32.40	0.957	0.054
Low + High	29.18	0.936	0.067
Mid + High	28.59	0.935	0.069

cosine similarity, a direct measure of the VLM's internal prediction confidence, likewise increases monotonically with altitude across all sequences without exception. The low absolute accuracy values across the board reflect the difficulty of zero-shot UAV semantic classification; the key finding is the within-sequence trend, not the absolute magnitudes.

TABLE IV
CLIP ZERO-SHOT SEMANTIC ACCURACY BY ALTITUDE TIER

Sequence	Tier	Accuracy (%)	Mean Cos. Sim.
HKisland01	Low	10.71	0.199
HKisland01	Mid	14.04	0.207
HKisland01	High	25.14	0.216
HKisland02	Low	5.68	0.200
HKisland02	Mid	14.05	0.204
HKisland02	High	22.65	0.215
HKisland03	Low	2.33	0.200
HKisland03	Mid	15.52	0.205
HKisland03	High	22.83	0.213
HKisland_GNSS03	Low	12.64	0.204
HKisland_GNSS03	Mid	15.38	0.212
HKisland_GNSS03	High	20.96	0.215

The mechanistic explanation is rooted in CLIP's fixed-resolution patch tokenization. CLIP ViT-B/32 divides each input image into a grid of 32×32 pixel patches, each encoded independently before cross-attention aggregation. At low altitude (20–28m), the effective ground sample distance is on the order of centimeters to decimeters per pixel, meaning that each

patch captures only a few square meters of surface texture—a granularity at which individual roof panels, road markings, or vegetation patches are visible, but no recognizable object boundary exists. This is fundamentally outside CLIP's pretraining distribution. At high altitude (58–63m+), a single patch subtends tens of meters, encompassing entire building footprints, vehicles, or tree canopies within a semantically coherent region. This is the scale at which CLIP's web-scale pretraining data was collected. Fig. 4 illustrates this scale dependence by overlaying CLIP cosine similarity maps for a representative semantic query at each altitude tier.

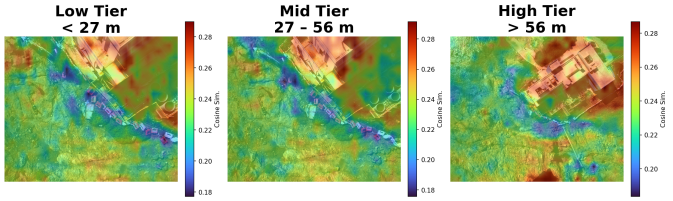


Fig. 4. CLIP cosine similarity maps for the query *"roof"* at (a) low, (b) mid, and (c) high altitude tiers from HKisland03. At low altitude, patches capture sub-object surface texture with no recognizable boundary, producing diffuse low-confidence activations. At high altitude, single patches encompass full building footprints, yielding coherent high-confidence activations.

C. Coverage Confound Analysis

A potential confound in the altitude-semantics relationship is geographic coverage: high-altitude images subtend larger ground areas and may incidentally capture more recognizable objects, inflating apparent semantic accuracy independent of perspective context. We directly test this hypothesis using the per-crop ground truth bounding box area logged during evaluation, which serves as a pixel-space proxy for object coverage under varying GSD conditions.

If the confound were genuine, larger bounding box areas—indicating objects occupying more pixels—should correlate positively with higher CLIP accuracy. The data demonstrates the opposite. As shown in Table V and Fig. 5, mean GT crop area monotonically decreases with altitude across all six sequences: from 2.89 Mpx at the low tier to 1.58 Mpx at the high tier, a 45% reduction. At the same time, accuracy increases from 11.7% to 22.4%. The correlation between bounding box area and accuracy is negative across all six datasets individually ($r = -0.225$ to -0.402), and highly significant in the global pool ($r = -0.311$, $p = 6.3 \times 10^{-53}$, $n = 1{,}558$ crops). Spearman's ρ confirms the result non-parametrically ($\rho = -0.188$ to -0.441, $p < 0.01$ in all cases).

These results establish that the accuracy improvement at high altitude is attributable to semantic context—the surrounding scene visible within a single image frame provides the object-level cues that CLIP's pretraining distribution requires—and not to incidental coverage area inflation. The confound runs in the direction opposite to the accuracy trend, making it a suppressor variable rather than an explanatory one.

TABLE V
COVERAGE CONFOUND REBUTTAL: ACCURACY VS. GT CROP AREA BY ALTITUDE TIER (GLOBAL MEAN ACROSS ALL 6 SEQUENCES, $n = 1{,}558$ CROPS)

Tier	Accuracy (%)	Mean GT Crop Area (Mpx)
Low	11.7	2.89
Mid	16.2	1.79
High	22.4	1.58

As altitude increases: accuracy ↑, crop area ↓
Pearson r(area, accuracy) $= -0.311$, $p = 6.3 \times 10^{-53}$

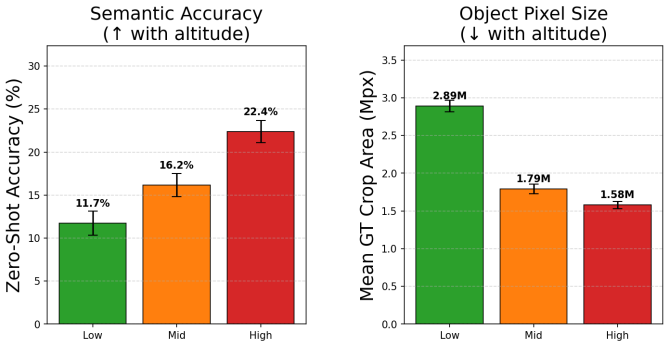


Fig. 5. Coverage confound rebuttal. Left: zero-shot accuracy increases with altitude tier. Right: mean GT bounding box area (pixel-space coverage proxy) *decreases* with altitude as growing GSD outpaces geographic coverage gains. The opposing trends falsify the hypothesis that accuracy gains are driven by larger object pixel representations.

D. The SAM 2 Benchmark Fallacy

Table VI reports SAM 2 Boundary F1 scores across all evaluated sequences and altitude tiers. Scores are uniformly and strikingly low across all conditions (F1 range: 0.045–0.140). A naive reading of these scores would suggest that SAM 2 performs poorly on UAV imagery. However, qualitative inspection reveals the opposite: SAM 2 accurately traces the fine-grained physical boundaries of objects—individual roof panel

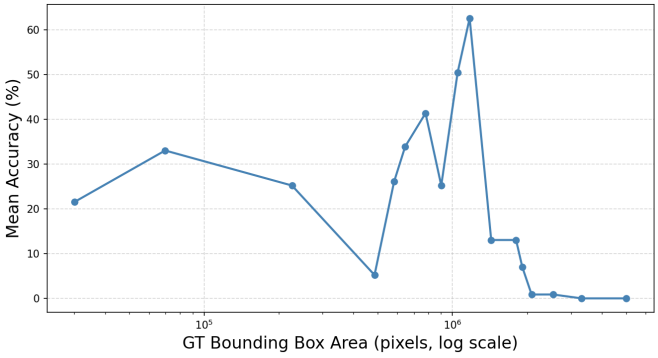


Fig. 6. Binned mean CLIP accuracy as a function of GT bounding box area (log scale, pooled across all sequences). The monotonically decreasing trend ($r = -0.311$, $p = 6.3 \times 10^{-53}$) confirms that accuracy improves despite objects occupying fewer pixels at high altitude, ruling out pixel-space coverage as the explanatory variable.

edges, vehicle contours, vegetation canopy outlines—while the ground-truth annotations are coarse human-drawn polygons with multi-meter pixel tolerance. The Boundary F1 metric, evaluated at a fixed pixel distance threshold, penalizes SAM 2 for producing accurate predictions that do not align with the deliberately simplified ground-truth boundaries. Fig. 7 illustrates this discrepancy directly.

TABLE VI
SAM 2 BOUNDARY F1 BY SEQUENCE AND ALTITUDE TIER

Sequence	Tier	Precision	Recall	F1
HKisland01	Low	0.041	0.221	0.069
HKisland01	Mid	0.027	0.136	0.045
HKisland01	High	0.053	0.155	0.079
HKisland02	Low	0.046	0.248	0.077
HKisland02	Mid	0.031	0.153	0.051
HKisland02	High	0.063	0.176	0.093
HKisland03	Low	0.042	0.232	0.071
HKisland03	Mid	0.032	0.169	0.054
HKisland03	High	0.057	0.160	0.083
HKisland_GNSS01	Low	0.028	0.148	0.047
HKisland_GNSS01	Mid	0.039	0.207	0.066
HKisland_GNSS01	High	0.086	0.249	0.127
HKisland_GNSS02	Low	0.027	0.151	0.046
HKisland_GNSS02	Mid	0.043	0.206	0.071
HKisland_GNSS02	High	0.097	0.258	0.140
HKisland_GNSS03	Low	0.030	0.167	0.051
HKisland_GNSS03	Mid	0.033	0.163	0.055
HKisland_GNSS03	High	0.076	0.206	0.111

Two structural patterns generalize across all six sequences. First, the mid tier consistently yields the lowest F1 in every sequence—lower than both low and high—because at intermediate altitude SAM 2 produces a moderate boundary density that is too fine to match coarse annotations but lacks the altitude-induced scale smoothing that reduces false detections at high altitude. Second, the consistently high recall confirms SAM 2 is finding real boundaries; precision fails only because the annotations do not annotate them. The standard UAV semantic evaluation protocol is therefore mathematically unsuited to evaluating modern foundation models. Any future benchmark for aerial scene understanding must either adopt fine-grained annotations at the precision level of the models being evaluated, or use annotation-agnostic metrics such as panoptic quality with relaxed boundary tolerance.

VI. LIMITATIONS AND FUTURE WORK

A. Trajectory Type and Tier Separation Quality

The altitude-accuracy relationship identified in Part I holds strongly for the non-GNSS spiral descent sequences (HKisland01–03, $r = 0.163$–0.203, $p < 0.001$), but is substantially weaker for the GNSS sequences ($r \approx 0.001$–0.078, $p > 0.10$). GNSS trajectories exhibit more aggressive heading changes and irregular altitude profiles during ascent and descent, which disrupt the clean tier separation achievable under controlled spiral descents. The current altitude-stratification approach assumes that altitude is the dominant source of variation between tiers; in practice, heading, velocity, and lighting also vary across a descent sequence, and their individual contributions to geometric and semantic performance remain unquantified.

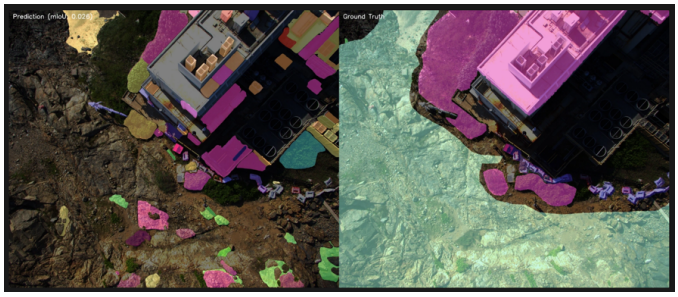


Fig. 7. SAM 2 predicted boundaries (left) and ground-truth polygon annotations (right) for a representative HKisland01 high-tier image. SAM 2 correctly traces sub-meter physical boundaries, while ground-truth annotations are coarse multi-meter polygons. Standard Boundary F1 metrics penalize the more accurate prediction for not aligning with the coarser annotation.

B. Geographic Coverage and Experimental Control

A methodological concern raised against this work is that the altitude-stratified evaluation is not a properly controlled experiment: as altitude increases, each image subtends a larger physical ground area, meaning the Low, Mid, and High tiers do not reconstruct the same scene at the same resolution. Under this criticism, any observed improvement in CLIP accuracy at high altitude could be attributed to the incidental capture of more complete objects across a larger geographic footprint rather than to any intrinsic property of altitude or perspective.

This concern is directly addressed by the coverage confound analysis in Section V-C. The ground-truth bounding box area—a pixel-space proxy for object coverage—monotonically decreases with altitude across all six sequences (from 2.89 Mpx at the low tier to 1.58 Mpx at the high tier, a 45% reduction), while accuracy simultaneously increases. The Pearson correlation between box area and accuracy is $r = -0.311$ ($p = 6.3 \times 10^{-53}$, $n = 1{,}558$ crops), confirmed non-parametrically by Spearman's ρ. If coverage were driving the accuracy trend, this correlation would be positive; it is strongly negative. The confound therefore acts as a suppressor variable rather than an explanatory one.

That said, a fully controlled experiment would fix the physical Ground Sample Distance (GSD) across tiers and evaluate reconstruction quality per unit ground area. UAVScenes does not provide calibrated intrinsic parameters in a form that permits rigorous per-image GSD normalization. Future work should report GSD explicitly, crop evaluation regions to a fixed physical footprint across altitude tiers, and evaluate the geometry-semantics tradeoff under matched-GSD conditions to further decouple altitude effects from resolution effects.

C. Perfect-Crop Assumption in CLIP Evaluation

The CLIP semantic evaluation uses ground-truth bounding boxes to crop object regions before encoding, providing an upper bound on zero-shot accuracy under perfect segmentation.

In a deployed pipeline, these crops would be produced by SAM 2, whose instance boundaries do not align with ground-truth class regions. The end-to-end SAM+CLIP pipeline (evaluate_sam_clip.py) was implemented but its full quantitative evaluation across all altitude tiers and sequences was deferred due to multi-day compute requirements per sequence. The gap between oracle-crop CLIP accuracy and real SAM-crop accuracy remains uncharacterized and is expected to be significant, particularly at low altitude where SAM 2 over-segments fine-grained surface texture.

D. Missing End-to-End Language Field Evaluation

The central motivation of Part I is the failure of language-embedded 3D representations on aerial data. However, the evaluation stops short of training LangSplatV2 on the altitude-stratified subsets and measuring open-vocabulary query accuracy in 3D, which would be the most direct test of the geometry-semantics paradox in the target application. This evaluation was deferred due to the multi-day per-sequence training cost of LangSplatV2. Future work should conduct this end-to-end evaluation—ideally across the Low+Mid combined split, which achieves the strongest geometric fidelity without the SfM failures of high-altitude training.

E. Annotation Quality for SAM 2 Evaluation

The SAM 2 benchmark fallacy identified in Section V-D renders the Boundary F1 metric uninformative for evaluating foundation model boundary quality against the UAVScenes annotations. This report identifies the problem and proposes annotation-agnostic alternatives, but does not implement them: relaxed-tolerance Boundary F1, panoptic quality, or instance-level IoU with SAM 2-generated pseudo-labels would all provide more meaningful assessments of SAM 2 performance on aerial data. Implementing these metrics is a direct next step.

F. Generalization Across Environments

All Part I experiments are conducted exclusively on Hong Kong urban island sequences from UAVScenes. The altitude-semantics relationship may differ in low-texture environments (open fields, water bodies) where CLIP's accuracy at high altitude is not aided by dense built-environment features, and in forest or vegetation-dominated scenes where both 3DGS and SAM 2 face different challenges. Validation on the AMtown and AMvalley sequences available in UAVScenes, or on datasets from structurally different environments, is needed to assess generalizability.

PART II: EFFECT OF MOTION BLUR ON 3DGS

VII. METHODOLOGY

A. Motion Blur Quantification

To mitigate the impact of motion-degraded frames on 3DGS optimization, we implement a blur quantification pipeline inspired by the frequency-domain analysis proposed by Masilamani et al. [5]. The algorithm evaluates the presence of high-frequency components in the image spectrum, which are characteristically suppressed during motion-induced blurring.

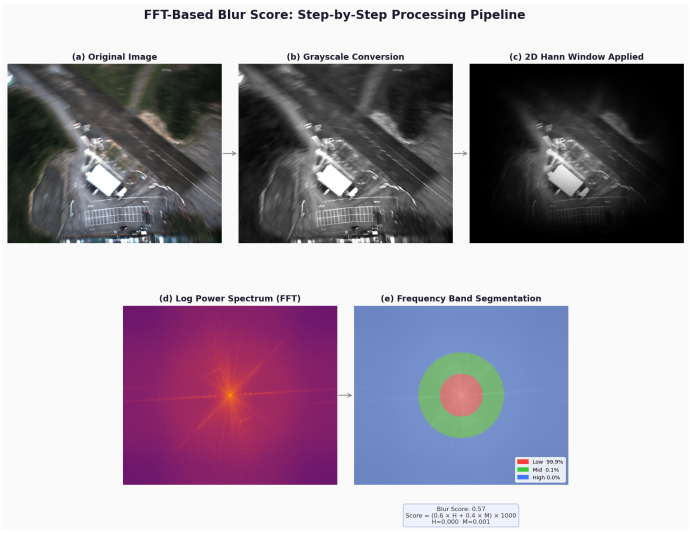


Fig. 8. Motion blur quantification pipeline with the FFT Algorithm

The input RGB image is first converted to a grayscale image to focus solely on structural information rather than chromatic variations. To minimize edge artifacts during the Fast Fourier Transform (FFT) process, a 2D Hann windowing function is applied. For an image of dimensions $H \times W$, the windowed image G_w is defined as:

$$G_w(y, x) = G(y, x) \cdot w(y, x) \tag{1}$$

where the 2D Hann window $w(y, x)$ is the outer product of two 1D Hann windows:

$$w(y, x) = \sin^2\left(\frac{\pi y}{H-1}\right) \sin^2\left(\frac{\pi x}{W-1}\right) \tag{2}$$

The windowed image is transformed into the frequency domain using a 2D FFT, followed by a quadrant shift to center the zero-frequency component. We compute the power spectrum $P(u, v) = |F(u, v)|^2$, where $F(u, v)$ is the discrete Fourier transform of G_w. For visualization and analysis of the dynamic range, the log power spectrum S_{log} is utilized:

$$S_{log} = \ln(1 + P(u, v)) \tag{3}$$

We segment the frequency domain into three radial regions of Low (L), Mid (M), and High (H), separated based on the Euclidean distance from the spectral center (u_c, v_c). Defining:

$$D(u, v) = \sqrt{(u - u_c)^2 + (v - v_c)^2} \tag{4}$$

the regions are partitioned by thresholds relative to the maximum image dimension D_{max}:

- **Low Frequencies:** $D(u, v) \leq 0.1 D_{max}$
- **Mid Frequencies:** $0.1 D_{max} < D(u, v) \leq 0.2 D_{max}$
- **High Frequencies:** $D(u, v) > 0.2 D_{max}$

The final blur metric, $\mathcal{B}$, is derived from the energy distribution across these bands. We calculate the energy ratios R_H

and R_M by normalizing the sum of power within the high and mid bands against the total spectral energy. The weighted blur score is then computed as:

$$\mathcal{B} = (0.6R_H + 0.4R_M) \times 1000$$

Higher values of $\mathcal{B}$ indicate a sharper image with preserved high-frequency details, while lower scores signify significant motion blur. This metric serves as the primary criterion for our subsequent threshold-based frame rejection strategy.

B. Thresholding and 3DGS

To isolate the impact of motion blur on novel view synthesis, we employ a thresholding strategy. While 3DGS relies on dense multi-view consistency, blurry images introduce corrupted gradients that degrade geometric and visual accuracy. Conversely, aggressively removing blurred images reduces view density, which can lead to sparse coverage. To systematically investigate this trade-off, we construct multiple training subsets based on varying blur score thresholds.

We start with the entire collection of captured images ($\mathcal{D}_{raw}$), where every individual image has been assigned a specific blur score ($S(I_i)$). We then establish a series of cut-off points, or thresholds ($\mathcal{T}$). For each cut-off point, we create a new, distinct subset of the data ($\mathcal{D}_j$). To build this subset, we look at every image in the original dataset and keep only those whose blur score is equal to or higher than the current cut-off point. Images with lower scores (i.e., the blurriest images) are discarded. By raising the threshold, we create increasingly smaller, but consistently sharper, training datasets. To guarantee that we can fairly compare the performance of models trained on these different subsets, we completely separate out a "test set" ($\mathcal{D}_{test}$). This test set contains only images that humans have manually confirmed are decently sharp, and these images are never included in any of the training subsets.

An aspect to note in this methodology is the deliberate exclusion of traditional Structure-from-Motion (SfM) pipelines for camera registration and initialization.

Typically, 3DGS relies on SfM to estimate where the cameras were when the photos were taken and to generate a starting point cloud. However, heavily blurred images lack the distinct, high-contrast features required by standard feature-matching algorithms. Attempting to run SfM on datasets containing severe motion blur results in failures.

To resolve the structural integrity of the reconstruction regardless of the blur threshold, we bypassed the SfM step entirely and utilized ground-truth positioning data provided from the UAVScenes dataset [11]:

- **Camera Poses:** The precise position and orientation of every camera were derived directly from high-precision Real-Time Kinematic (RTK) GPS data recorded during the capture process.
- **Gaussian Initialization:** Rather than relying on an inherently noisy, SfM-generated sparse point cloud, the 3D Gaussians were initialized using corresponding ground-truth LiDAR point cloud data.

This would imply that geometric degradation observed in the final 3DGS models are the result of the blur present in the training images, rather than a secondary failure caused by inaccurate camera poses.

VIII. Experimental Setup

A. Dataset

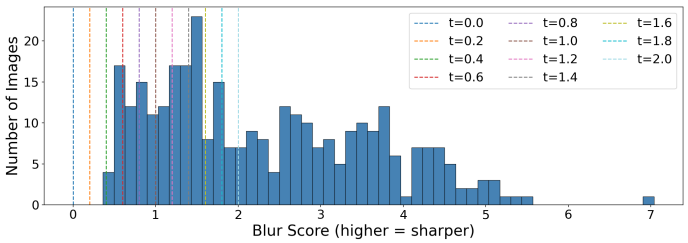


Fig. 9. Blur Score distribution of selected dataset

For empirical evaluation, a selected sequence from the 'HKAirport_GNSS_Evening' dataset was utilized. The selected subset comprises 336 frames, spanning the timestamp indices 1698227516.600 to 1698227548.999. From this selection, 10 images were held out exclusively to serve as the test set. Figure 9 illustrates the distribution of calculated blur scores across the raw dataset, alongside the discrete thresholds applied for subset generation.

B. Evaluation Metrics

To quantitatively evaluate the visual fidelity of the reconstructed 3D Gaussian splats, we adopt the standard evaluation metrics established in the original 3DGS framework [2] which are PSNR, SSIM and LPIPS as explained in section IV subsection B. The rendered novel views are compared against the held-out ground truth test set $\mathcal{D}_{test}$.

IX. Results

TABLE VII
Quantitative evaluation of 3DGS reconstruction fidelity across varying blur rejection thresholds.

Threshold	PSNR ↑	SSIM ↑	LPIPS ↓
0.0	20.452	0.632	0.556
0.2	20.536	0.633	0.556
0.4	20.489	0.634	0.559
0.6	20.576	0.636	0.551
0.8	20.615	0.639	0.548
1.0	20.557	0.637	0.544
1.2	20.712	0.641	0.536
1.4	21.005	0.652	0.514
1.6	21.069	0.659	0.509
1.8	21.188	0.663	0.495
2.0	21.568	0.674	0.477

The blur score thresholds were set such that they range from a baseline of $\tau = 0.0$ (no rejection) to a strict threshold of $\tau = 2.0$ in intervals of 0.2. The quantitative performance of these models, evaluated against the held-out sharp test set using PSNR, SSIM, and LPIPS, is summarized in Table VII.

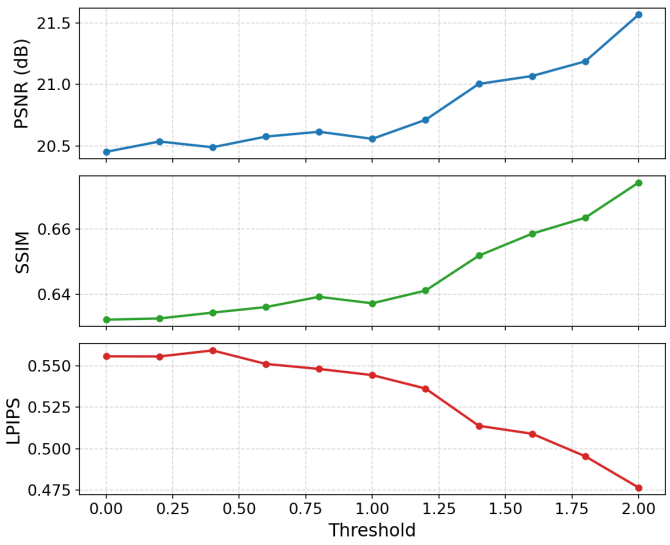


Fig. 10. 3DGS reconstruction fidelity across varying blur rejection thresholds

The data reveals a clear, positive correlation between the strictness of the blur rejection threshold and the final fidelity of the novel view synthesis. Across all three metrics, the baseline model containing the full, unaligned dataset ($\tau = 0.0$) performed the poorest, yielding a PSNR of 20.452 dB and an LPIPS of 0.556. Conversely, the model trained on the most aggressively filtered dataset ($\tau = 2.0$) achieved the highest performance across all categories (PSNR: 21.568 dB, SSIM: 0.674, LPIPS: 0.477).

These results are quite interesting. In the lower threshold ranges ($\tau = 0.0$ to 1.0), performance improvements are quite minimal and occasionally oscillatory. For instance, increasing the threshold from 0.0 to 1.0 only improves PSNR by 0.105 dB and SSIM by 0.005. This suggests that while some highly blurred images are removed, the remaining moderately blurred images still introduce sufficient noisy gradients to bottleneck the 3DGS optimization process.

However, a distinct inflection point occurs at $\tau = 1.2$. Beyond this threshold, the metrics exhibit a sharp, monotonic improvement. Notably, the LPIPS score, which serves as the strongest indicator of perceptual human realism, exhibits its steepest drop in this range, falling from 0.536 at $\tau = 1.2$ to 0.477 at $\tau = 2.0$.

This suggests that the largest improvements in 3DGS fidelity do not occur with the removal of the bottom most blurriest images, but at a critical threshold defined for each dataset. A possible explanation to this is that when an image is extremely blurred, the structural edges are completely obliterated, reducing the image to a low-frequency image of larger smeared patches. During the pixel-wise gradient descent ($\mathcal{L}_1$ loss), the optimizer essentially sees this as uniform background noise. Because it lacks any strong, defined edges, it exerts a very weak, diffuse pull on the Gaussians. However, a moderately blurred image still contains distinct, high-contrast edges, but those edges are smeared or shifted by a few pixels across the axis of motion. To the optimizer, this looks like valid geometric data such that when it compares a sharp view of an edge against a moderately blurred view of the same edge, it attempts to satisfy both. This active conflict forces the densification process to spawn unnecessary Gaussians near the actual surfaces.

X. Limitations and Future Work

While the proposed FFT-based thresholding methodology successfully isolated the impact of motion blur on 3DGS reconstruction, the current study operates under several constraints that require further investigation.

A. Sparsity Tradeoff

Firstly, the experimental design hypothesized an inverted U-curve relationship between the rejection threshold and final rendering fidelity. Intuitively, the fidelity should initially rise as blurry artifacts are removed, then fall as the dataset becomes too sparse to support coherent 3D geometry. However, our maximum evaluated threshold ($\tau = 2.0$) only captured the ascending portion of this curve. Because we did not raise the threshold further, we did not observe the anticipated degradation in 3DGS quality due to data sparsity.

B. Scope of Evaluation

Due to computational and temporal constraints, the quantitative analysis was restricted to a single continuous sequence from the dataset ('HKAirport_GNSS_Evening'). While this sequence provided a highly controlled environment for evaluating the thresholding pipeline, it limits the immediate validation of the methodology across diverse environments, varying camera trajectories, and different lighting conditions.

C. Scene Dependence

A critical limitation of the proposed FFT-based blur metric lies in its reliance on absolute energy ratios. Empirical observations during secondary testing revealed that while the metric is highly effective at relatively sorting images by blur severity within a specific sequence, the absolute numerical scores are intrinsically scene-dependent. The baseline high-frequency distribution of an image is heavily dictated by its natural semantic content. For example, a scene dominated by smooth gradients (e.g., human architectures) inherently possesses significantly less high-frequency energy than a highly textured environment (e.g., dense forests).

Because of this scene-dependent frequency bias, the absolute blur scores calculated cannot currently be used as a universal threshold across unseen datasets. A threshold of $\tau = 2.0$ might effectively isolate sharp images in an urban dataset but could erroneously discard perfectly sharp images in a low-texture environment. Future work must address this limitation by developing a scene-agnostic, normalized scoring mechanism.

XI. Discussion and Conclusion

This report presents empirical evidence for two independent challenges that compound to prevent current aerial spatial AI systems from achieving high-quality, semantically rich 3D representations in practice.

Part I establishes a fundamental geometry-semantics paradox across a single UAV trajectory: the altitude that maximizes 3DGS fidelity (low, PSNR $\sim$33.7 dB) minimizes VLM semantic accuracy (low, $\sim$11.7%), while the altitude that enables coherent semantic understanding (high, $\sim$22.4% accuracy) systematically causes SfM registration failures and PSNR degradation to $\sim$26.2 dB. Critically, this accuracy trend is not a coverage artifact—GT bounding box area decreases by 45% as altitude increases ($r = -0.311$, $p = 6.3 \times 10^{-53}$), ruling out pixel-space object size as an explanatory variable and confirming that the improvement is attributable to semantic context alignment with CLIP's pretraining scale. Additionally, the uniformly low SAM 2 Boundary F1 scores (0.045–0.140) are shown to be a benchmark artifact of annotation coarseness rather than model failure, with recall values (0.136–0.258) consistently indicating correct detection at sub-annotation granularity.

Part II demonstrates that motion blur represents a second independent bottleneck on 3DGS fidelity. The FFT-based blur quantification pipeline successfully ranks images by blur severity, and progressive threshold-based frame rejection yields monotonic improvements above a scene-specific inflection point ($\tau = 1.2$), achieving gains of up to 1.1 dB PSNR, 0.04 SSIM, and 0.08 LPIPS. The mechanistic explanation—that moderately blurred frames inflict greater optimization damage than severely blurred ones—is supported by the oscillatory plateau below the inflection point and the sharp improvement above it.

Together, these findings motivate a unified design principle for future aerial spatial AI systems: trajectory design must decouple geometry and semantics through multi-tier flight profiles that physically separate low-altitude reconstruction passes from high-altitude semantic survey passes, and data preprocessing must enforce blur-aware curation before any reconstruction pipeline is invoked. Both interventions are tractable with current hardware and represent concrete next steps toward language-queryable 3D maps from UAV platforms.

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